Programming Questions- Chapterwise

Loops + Conditionals

1. WAP to check even/odd.
2. WAP to check the number is +ve/-ve/0.
3. WAP to check if a character is a vowel or not.
4. WAP to print N natural number.
5. Find Largest/Smallest of two numbers.
6. Find Largest/Smallest of three numbers.
7. WAP to print sum of N natural numbers.
8. WAP to find the factorial of a given number.
9. WAP to check if the number is in range or not like 0 to 100, -50 to 70.
10. WAP to implement a basic calculator(switch case).
11. WAP to print 2 power series.
12. WAP to print x^y.
13. WAP to find combination nCr.
14. WAP to check if a year is leap or not.
15. WAP to check if a number is prime or not.
16. WAP to check if a 3-digit number is armstrong or not.
17. WAP to check if any digit number is armstrong or not.
18. WAP to check if a number is strong or not.
19. WAP to check if a number is perfect or not.
20. WAP to print prime up to limit L.
21. WAP to print N prime numbers.
22. WAP to print positive fibonacci series.
23. WAP to print negative fibonacci series.
24. WAP to print ap and gp series.
25. WAP reverse an integer number.
26. WAP to check if the number is palindrome.
27. WAP to count digits of a given integer.
28. WAP to remove 0 from a given integer.
29. WAP to find the sum of digits of a given number.
30. WAP to find LCM of two numbers.
31. WAP to find GCD/HCF of two numbers.
32. WAP to swap two integers.
33. Find the largest of N elements without using an array.
34. Print N natural number without using loop statement.
35. Print the table of a given number up to the limit ‘L’ , like 3 x 1 = 3, 3 x 2 = 6.
36. Print at least 20-30 patterns.
37. Do some questions using the ternary operator.
38. Swap two values without using a third variable.
39. Find the square root of an integer correct up to 2 decimal places.
40. Print Square ring pattern.
41. Print pascal’s triangle.

1-D Array

1. Find sum/avg of elements of given array.
2. Search one element in the array (available/not).
3. Print array elements in reverse order.
4. Find Largest and Smallest of the array.
5. Reverse the array.
6. Reverse each element of the array at their index.
7. Sort array in ascending/descending order using bubble sort.
8. Shift all 0's to the left side of the array keeping other's order the same.
9. Check array is palindrome or not.
10. Check array is unique or not.
11. Print only unique elements in the array.
12. Print only repeated elements in the array.
13. Print frequency of each element of array.
14. Find union of two arrays.
15. Find Intersection of two arrays.
16. Merge second array into first array.
17. Remove duplicates from the array and create a new array.
18. Print array without repetition.
19. Convert array elements as 0/1 based on even/odd and find the decimal value of the array.
20. Rotate array N times by right/left based on +ve/-ve entry from user.
21. Find possible pairs from the array whose sum is given by the user.
22. Find the largest sum pairs in the array.
23. Find the Power set of a unique array. [2^n-1].
24. Find the Nth Largest/Smallest element of the array.

Function and pointer basics

1. WAP to show the difference between Pass by value and reference.
2. Print array using function.
3. WAP to swap two values of any type.
4. Check system is little/big endian.
5. Convert little to big endian system.
6. Find the size of the variable without using the sizeof() operator.
7. Try array/strings problem using function.

Recursion

1. WAP to print “Hello” n times.
2. WAP to print n natural numbers (both sides).
3. WAP to print reverse n natural numbers.
4. WAP to find the sum of n natural numbers.
5. WAP to find factorial.
6. WAP to find x^y using recursion.
7. WAP to print 2 power series using recursion.
8. WAP to print positive fibonacci series.
9. WAP to print reverse n natural numbers except prime numbers.
10. WAP count number of digits of integer.
11. WAP to find the sum of digits of a given number.
12. WAP to reverse array.
13. WAP to reverse the string.
14. WAP to print binary using recursion.
15. WAP to convert Decimal to Hexadecimal using recursion.

Strings

1. WAP to print the string using the loop.
2. WAP to print the string using the function.
3. WAP to reverse the string.
4. WAP to count the number of vowels in the string.
5. WAP to check if the string is palindrome or not.
6. WAP to check if a string is unique or not.
7. WAP to convert multiple spaces into single spaces of the string.
8. WAP to remove duplicates.
9. WAP to validate a file name is with extension ‘.c’ or not.
10. WAP to remove all characters of the second string into the first string.
11. WAP to check if a string is unique or not.
12. WAP to print frequency of each character of string.
13. WAP to print only repeated characters of the string.
14. WAP reverse the string word wise.
15. WAP to reverse each word of the string at their fixed place.
16. WAP to implement atoi().
17. WAP to implement itoa().
18. WAP to implement strlen();
19. WAP to implement strcpy().
20. WAP to implement strcmp().
21. WAP to implement strcat().
22. WAP to implement strchr().
23. WAP to implement strstr().
24. WAP to implement strtok().
25. WAP to take a binary(0/1) string and print its decimal value.
26. WAP to print all the smallest word of string (by length).
27. WAP to print all the longest word of string (by length).
28. WAP to print each word of string (both sides).

29. WAP to check if the string is pangram or not.
30. WAP to take the string like “Q2ero4wertg1K” to give output like “erwertK”.
31. WAP to validate password string with one capital, small, digit, special and length be 8 at least.
32. WAP to validate the IP address of ipv4 with its property.
    Syntax : **xxx.xxx.xxx.xxx ,** where each part is 0 - 255 (number only).

33. WAP to validate the IP address of ipv6 with its property.
    Syntax : **xxxx.xxxx.xxxx.xxxx.xxxx.xxxx.xxxx.xxxx,** where each part is 4 digits Hexadecimal only.
34. WAP to reverse the IPv4 address like “123.34.232.10” -> “10.232.34.123”.

Bitwise OPerators

1. Check number is even/odd.
2. Check number is 2’s power or not.
3. Find Double/Half of the given number.
4. Swap two integers.
5. Clear 1st set bit from LSB.
6. Count the number of set bits.
7. Print 2 power series.
8. Print binary of given integer.
9. Make number -ve without using ‘-’ sign.
10. set/get/clear/toggle a bit at a particular index position of a given number.
11. Print n bits from LSB.
12. Get n bits from LSB and print value.
13. Reverse the binary of a given number using a new number.
14. Implement post inc/dec operator using bitwise (as a function).
15. Set/get/toggle/clear n bits at given position.
16. Swap nibble of a byte.
17. Apply circular left/right shift count times.
18. Find the XOR value of two numbers without ^ operator.
19. Find the only non-repeating number in the array.
20. Find the missing number in the group of n natural numbers.
21. WAP to print the decimal value of each byte of a given integer from the LSB side.

FileIO

1. Print the data of .txt file
2. Create a copy of 1.txt as 2.txt file.
3. Print the content of the .txt file in reverse order.
4. Find the size of the given file.
5. Count the number of words in the file.
6. Count the number of lines in the file.
7. Print the nth line of the file.
8. Print Last Line of the file.
9. Scan n integers from user and store in .txt file (each in new line).
10. Scan n names from user and store in .txt file (each in new line).
11. Delete the nth line from the given .txt file.
12. Print the longest line of the .txt file.
13. Remove all single line comments from the given .c file.

2D array

1. scan and print 2D array elements.
2. find the avg of 2D array elements.
3. print nth row of 2D array.
4. print nth col of 2D array.
5. find the largest element of the whole 2D array.
6. find the largest element of each row.
7. find the largest element of each column.
8. print transpose of 2D array using func (without extra space).
9. find the sum of 2 matrices (both dim exactly the same).
10. Find the multiplication of 2 matrices (where c1=r2).
11. print both diagonal of 2D array in X pattern format.[square type]
12. print both diagonal element normally: [sq type]
13. find the sum of both diagonal elements.
14. WAP to swap both diagonal elements. [sq type]
15. reverse each row of the 2D array.
16. reverse each col of the 2D array.
17. sort each row of 2D array(asc/des).
18. rotate the 2D array by 90 degree(clockwise).
19. rotate the 2D array by 90 degree (anticlockwise).
20. update the 2D array elements with just the next prime, if it is not prime.

Advance pointers

Macro

UDT [structure, union]